MATH 122 — Calculus II · Exam 3: Sequences and Series

Name: _____ **Date:** _____

Instructions:

- Show all work: Full credit requires complete, correct steps and proper math notation.
- Points: Each question is worth 10 points.
- Formula Sheet: Permitted (formulas only; no worked examples).
- Calculators: Basic, non-graphing calculators allowed.

Part I: Limits of Sequences

1. (10 points) Use L'Hôpital's Rule to determine whether the sequence converges. If it converges, find its limit.

$$\left\{\frac{\ln n}{\sqrt[3]{n}}\right\}_{n=1}^{+\infty}$$

2. (10 points) Use the Squeezing Theorem for Sequences to find the limit.

$$\lim_{n\to+\infty} \frac{n+\cos n}{4n+3}$$

3. (10 points) Use the Squeezing Theorem for Sequences to find the limit.

$$\lim_{n\to+\infty} (2^n + 7^n)^{1/n}$$

(Hint: compare $2^n + 7^n$ with 7^n and with $2 \cdot 7^n$.)

4. (10 points) Assume the sequence below converges and all of its terms are positive. Find its limit.

$$a_1 = 1, \qquad a_{n+1} = \frac{1}{2}\left(a_n + \frac{7}{a_n}\right)$$

Part II: Monotone Sequences

5. (10 points) Use the difference $a_{n+1} - a_n$ to show that the sequence is strictly increasing or strictly decreasing.

$$\left\{\frac{n}{4n+1}\right\}_{n=1}^{+\infty}$$

6. (10 points) Use the ratio a_{n+1}/a_n to show that the sequence is strictly increasing or strictly decreasing.

$$\left\{\frac{n}{4^n}\right\}_{n=1}^{+\infty}$$

Part III: Infinite Series

7. (10 points) Determine whether the geometric series converges. If it converges, find its sum.

$$\sum_{k=0}^{\infty} \frac{(-1)^k\, 4}{3^k} = 4 - \frac{4}{3} + \frac{4}{3^2} - \cdots + \frac{(-1)^k\, 4}{3^k} + \cdots$$

8. (10 points) Determine whether the geometric series converges. If it converges, find its sum.

$$\sum_{k=2}^{\infty} \frac{2^k}{3^{k-1}} = \frac{4}{3} + \frac{8}{9} + \frac{16}{27} + \cdots + \frac{2^k}{3^{k-1}} + \cdots$$

9. (10 points) Find a closed form for the nth partial sum s_n, then determine whether the series converges. If it converges, find its sum.

$$\sum_{k=1}^{\infty} \frac{1}{k^2+5k+6} = \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \cdots + \frac{1}{k^2+5k+6} + \cdots$$

10. (10 points) Find a closed form for the nth partial sum s_n, then determine whether the series converges. If it converges, find its sum.

$$\sum_{k=1}^{\infty} \frac{3}{k(k+3)} = \frac{3}{1\cdot 4} + \frac{3}{2\cdot 5} + \frac{3}{3\cdot 6} + \cdots + \frac{3}{k(k+3)} + \cdots$$